

“I Thought It Was Private”: User Misconceptions and Risk Perception in AI Assistant Use

Raameen Ahmed, Pedro Callado de Paiva, Muhammad Sheraz, Mohammed Alsheraa, Lea Nahli
University of Michigan-Dearborn, {raameena, ppaiva, msheraz, shewa, nlea}@umich.edu

Abstract—AI assistants are widely used, despite limited user understanding of how they handle sensitive data, potentially creating risks of overtrust and unsafe disclosure. This study investigates the relationship between user misconceptions, trust, and risky behavior when interacting with AI systems. We conducted a survey of about 64 participants, measuring objective knowledge, perceived understanding, trust, and data-sharing behaviors, along with a short educational intervention.

Although 73.8% of participants correctly answered all knowledge-based questions, many reported low confidence in understanding data handling and still engaged in risky behaviors. Frequent users were significantly more likely to share sensitive information (63%) compared to less frequent users (14%), despite reporting low trust in AI systems. Statistical analysis showed that both trust and usage patterns are associated with information sharing behavior, highlighting a disconnection between user awareness, perception, and action.

Additionally, a brief educational intervention significantly improved verification behaviors for both text ($p = 0.0157$) and links ($p = 0.0003$). Our findings demonstrate that risky AI use is not driven solely by a lack of awareness, but by a gap between knowledge and behavior, and that targeted education can effectively improve safer practices.

Index Terms—AI Trust, Privacy Behavior, Risk Perception, Misconceptions, Data Sharing Behavior, Security Awareness

I. INTRODUCTION

In the advent of Artificial Intelligence (AI), AI-driven assistants have become an extension of many people’s digital undertakings, ranging from casual use to complex and important tasks. As the technology further develops and becomes more integrated into modern systems, their perceived reliability increases. Users feel less like they are issuing commands to a tool and more like they are speaking to an intelligent partner. As the integration of AI grows, so does user reliance on it, often without an understanding of its limitations, data handling practices, or potential risks.

Alongside the increasing use and adoption of AI assistants is the growing concern that users may overtrust them. Despite the information never being fully guaranteed for its effectiveness, too much trust in an AI assistant may lead a user to accept outputs assuming they are accurate and safe. Overreliance can lead to problematic behaviors, such as sharing sensitive personal information, accepting incorrect responses, or failing to verify outputs.

Overreliance finds its origins in user misconceptions [1]. Many users do not fully understand how AI systems generate responses, what data is kept by the AI assistant, or how their inputs may be used by the systems for internal operations such as training [2]. These misconceptions about what happens

to user data may contribute to privacy risks and sensitive data being exposed to AI assistants. In some cases, a user may expose their personal, financial, or work-related data to an AI chatbot, assuming that it can be trusted and ensures confidentiality.

Our study investigates the overtrust users may have in AI assistants and examines user behavior when it comes to handling sensitive data when conversing with an AI system. This research aims to find patterns in user beliefs, evaluate how aware users are of information they put into AI systems, and analyze the relationship between how trusting a user is of AI and the risky behaviors they engage in when interacting with AI. To achieve these ends, we formulated a survey of 20 questions and administered it to 40 participants. The survey was structured around behavior, frequency of AI use, and handling of sensitive data. This will allow us to gain greater insight into dangerous habits when interacting with AI, trust levels, and user misconceptions.

Through this research, we can understand the origins of what makes users overtrust AI assistants. This research becomes valuable information for designing safer systems, improving transparency, and correcting user behavior to help users make more educated decisions when interacting with AI.

II. BACKGROUND & RELATED WORK

A. Background

The rapid integration of Large Language Model (LLM) chatbots into daily life has fundamentally changed how users interact with technology [3]. Unlike traditional software, AI assistants are designed to be conversational, personable, and engaging, traits that encourage users to share information freely [3]. Recent estimates indicate that OpenAI serves over 800 million users weekly, with approximately 18 billion messages exchanged per week [3]. Among these users, roughly 30% of consumer ChatGPT usage is tied to work-related tasks, meaning millions of users are feeding these systems commercially sensitive, personally identifiable, or otherwise private information [3].

The risks associated with this widespread adoption are not merely theoretical. Research has documented that 43% of employees have shared sensitive information, including financial and customer data with AI tools [4]. More alarmingly, 38% of employed AI users admitted to submitting sensitive work-related information to AI tools without their employer’s knowledge, with Gen Z and Millennials being the most likely

to engage in this behavior [5]. As one researcher noted, "While executives debate AI strategy in boardrooms, 93% of their employees have already made the decision for them, and they're sharing confidential data with unapproved AI tools" [4].

This disconnection between organizational preparedness and user behavior creates what researchers have termed a "perfect storm of data security risks, compliance violations, and customer trust issues" [4]. However, despite growing awareness of technical vulnerabilities, comparatively little is known about the human factors driving these behaviors specifically, what users believe about AI data practices and how those beliefs influence their willingness to share sensitive information.

B. Related Work

Prior research on AI privacy and security has developed along three primary trajectories: (1) technical demonstrations of vulnerabilities, (2) user perception and behavior studies, and (3) trust and overtrust in human-AI interaction. We review each below and identify the critical gap our work addresses.

1) Technical Vulnerability Research: A substantial body of work has documented the technical mechanisms by which AI assistants can be manipulated or compromised. Zhan et al. [6] demonstrated that malicious actors can transform benign conversational AI systems into effective data-harvesting agents through nothing more than carefully crafted system prompts. In a randomized controlled experiment with 502 participants, they found that malicious CAIs elicited personal information from over 90% of participants, compared to just 24% in baseline conditions. Critically, their "reciprocal" strategy, which combined helpful responses with empathy and emotional support, proved particularly effective at inducing disclosure without raising users' privacy risk perceptions.

Concurrently, Check Point Software Technologies [7] identified a DNS tunneling vulnerability in ChatGPT's code execution environment that could allow attackers to exfiltrate sensitive data, including uploaded files and the model's own analytical summaries, without triggering security alerts. This research revealed that even when users believe their data remains private within a "sealed workspace," technical vulnerabilities can enable silent data leakage.

In the realm of agentic AI systems, Zhang et al. [8] demonstrated that LM agents acting on users' behalf can inadvertently leak private information in interpersonal communication tasks. Their study found that when users were given the choice between their own drafted responses and agent-generated responses, many preferred the agent's response even when it contained more privacy violations, leading to an increase in harmful disclosure from 15.7% to 55.0%.

2) User Privacy Perception and Behavior Research: Several recent studies have examined how users perceive and navigate privacy risks in LLM-based systems. Gumusel et al. [9] surveyed 267 U.S.-based LLM users and found low awareness of privacy policies, moderate concern over data handling, and reluctance to share the most sensitive information (e.g., social security numbers). However, they also

observed that usage frequency and prior experience strongly influenced comfort levels, suggesting that familiarity may breed complacency.

Malki et al. [10] conducted a survey of 211 UK-based conversational AI users and identified a critical finding: many participants held inaccurate beliefs about the effects of deleting data and opting out of model training. This suggests that users operate under flawed mental models of how AI systems store and use their information, a finding consistent with earlier work on privacy in other digital domains but newly applied to the AI context.

Industry surveys have reinforced these academic findings. A Concentrix study [11] of 1,000 U.S. adults found that trust patterns vary significantly across generations: Gen Alpha (ages 10-20) "won't 'adopt' AI, they'll assume it," while Millennials (20-40) value efficiency and privacy, Gen X (40-60) prioritize reliability, and Boomers (60+) "trust humans more than systems."

3) Trust and Overtrust in Human-AI Interaction: The phenomenon of overtrust defined as "misunderstanding the risks associated with an action" has emerged as a central concern in human-AI interaction research. Yazan et al. [12] surveyed 173 users about their preferences between ChatGPT and Google for search tasks, identifying two distinct user groups. The "dual-user" group (using both ChatGPT and Google daily) exhibited higher trust in ChatGPT, perceived it as more human-like, and expressed willingness to trade factual accuracy for enhanced personalization and conversational flow. This finding aligns with Cheng et al.'s [13] demonstration that AI models affirm user behavior 49% more frequently than humans do, even in scenarios involving deception or harm. Critically, Cheng et al. [13] found that users rated sycophantic (overly agreeable) responses as more helpful and trustworthy, and expressed greater willingness to rely on such systems again meaning the very feature that causes harm also drives user engagement.

Yoon's doctoral dissertation [14] provides a comprehensive theoretical framework for understanding the evolution from trust to overreliance in human-AI interactions, applying socio-technical and psychological frameworks to explain how AI features and user sociability influence perceptions of interaction quality and trust formation.

C. The Research Gap: What is Missing

Despite the substantial body of research summarized above, a critical gap remains:

What Existing Research Has Established:

- Technical attacks can manipulate AI assistants [6]
- Users have inaccurate beliefs about data retention, its deletion, and opt-out [10]
- Users overtrust AI and exhibit sycophancy preference [13]
- There are demographic patterns in trust levels [11]
- Privacy awareness is low [9]

What We Still Do NOT Know (Our Contributions):

- Whether users understand these attacks are possible

- The full range of misconceptions users hold about AI data practices
- How specific misconceptions map to specific risky behaviors
- Which user groups are most susceptible to different types of risks
- Whether education about risks changes actual behavior

Critical observation: Most existing research is technically driven (demonstrating what is possible) or descriptive (measuring awareness levels). Few studies have attempted to:

- Systematically map user misconceptions to specific technical vulnerabilities
- Measure whether targeted education changes risk perception and behavior
- Bridge the gap between security research and human-computer interaction in the AI context

D. Our Contributions (Novelty Statement)

Based on the gap identified above, our work makes the following novel contributions:

- **User- and behavior-driven focus:** Unlike prior work centered on technical vulnerabilities, we center the human user as the primary unit of analysis, examining not just what users know but what they believe and how those beliefs potentially drive behavior.
- **Systematic misconception measurement beyond awareness:** While prior studies have measured privacy awareness levels, we develop a structured instrument to identify specific, actionable misconceptions about data storage, usage, retention, and exposure risks.
- **Mapping misconceptions to risky behaviors:** We go beyond correlation to examine how specific misconceptions (e.g., "deleting my chat removes my data") predict specific risky behaviors (e.g., pasting confidential documents).
- **Education intervention testing:** We assess whether a targeted educational intervention explaining real attack mechanisms in accessible language changes risk perception and behavioral intentions, providing evidence for practical intervention design.
- **Bridging security + HCI:** We integrate insights from technical security research (attacks and vulnerabilities) with human-computer interaction methods (survey design, user profiling) to address a problem that neither field can solve alone.

III. METHODOLOGY

A. Study Design

This study adopts a quantitative, survey-based methodology to investigate user awareness, trust, and behavior when interacting with AI assistants. The study consists of two main components: (1) a cross-sectional survey measuring user knowledge, perceptions, and behaviors, and (2) a short educational intervention designed to assess whether increased awareness leads to measurable changes in reported behavior.

The survey was structured to capture relationships between user misconceptions, trust in AI systems, and engagement in potentially risky behaviors. A post-intervention component was included to evaluate changes in user responses after exposure to targeted educational content.

B. Participants and Recruitment

A total of 64 participants were recruited using convenience sampling. The survey was distributed through multiple informal channels, including student group chats (e.g., WhatsApp, email) and in-person outreach. To reduce bias from prior exposure to security-specific content, we intentionally excluded participants from the course in which this study was conducted.

The participant pool primarily consisted of university students from a variety of institutions, such as the University of Michigan-Dearborn and Berklee College of Music, and academic disciplines, including but not limited to engineering, business, and biological sciences. Some responses were also collected from non-student participants through personal outreach (e.g., family members). No formal inclusion criteria were imposed, and participation was entirely voluntary and anonymous. No further incentives were provided.

Although the sample was primarily composed of university students, which may limit our ability to generalize to broader populations, this demographic represents a high-frequency user group of AI assistants, making their behavior particularly relevant to this study.

C. Survey Instrument

The survey consisted of 21 questions organized into the following logical categories:

- **Demographics:** Basic participant characteristics (e.g., age group, education level).
- **AI Usage:** Frequency and purpose of AI assistant use.
- **Knowledge & Misconceptions:** True/false and conceptual, open-ended questions assessing understanding of AI data practices and system behavior.
- **Behavior:** Self-reported actions related to information sharing and verification of AI-generated content.
- **Risk Perception & Trust:** Participant self-reported trust in AI and assessment of the risks associated with specific usage scenarios.
- **Education Impact:** Short post-intervention questions measuring changes in behavior and perception.

The survey included a mixture of question types, including multiple-choice (single and multiple selection), Likert-scale questions (1-4 and 1-5 point scales), numerical input (e.g., estimated frequency of AI use), true/false questions, one ranking question, and two open-ended responses.

D. Educational Intervention

An educational intervention was incorporated into the survey to assess whether increased awareness influences user behavior. The intervention consisted of a structured informational section comprising 20 concise bullet points. The first

portion focused on correcting common misconceptions about AI systems, including data storage and logging practices, the potential reuse of user inputs for training or review, the persistence of data across sessions, and the risks associated with prompt injection and external integrations. It also emphasized that AI outputs are not guaranteed to be accurate or secure and may expose sensitive information.

The second portion provided actionable guidance on safer AI usage practices and behaviors. These included avoiding the disclosure of sensitive personal or professional information, redacting or anonymizing data before providing it as input, verifying AI-generated outputs using trusted sources, being cautious with AI-generated links or commands, and consulting qualified professionals for relevant decisions. Our information-based intervention explicitly encouraged participants to treat AI systems as tools rather than trusted authorities and to assume that shared data may be stored or reused beyond the immediate interaction, concluding that it is their sole responsibility to verify and protect their data.

This information-based intervention was presented after the initial survey questions and before the final Education Impact section. Following exposure to this content, participants were asked a subset of revised or repeated questions to test whether correcting misconceptions leads to measurable improvements in safer AI-use behavior and perception.

E. Measures & Variables

This study defines several key constructs:

- **Awareness:** Assessed through responses to knowledge-based questions regarding AI data handling, storage, and reliability. Correct responses were interpreted as indicators of higher awareness.
- **Trust:** Measured using Likert-scale responses indicating the degree to which participants trust AI systems to handle their data securely.
- **Behavior:** Evaluated through self-reported actions, including the types of information shared with AI systems and the frequency of verifying AI-generated outputs.
- **Risk Perception:** Measured using multiple-choice questions and scenario-based questions in which participants rated the perceived risk of specific AI-related actions and sharing behavior.
- **Education Impact:** Assessed by comparing pre- and post-intervention responses to questions related to verification behavior and risk awareness.

F. Data Analysis

All data analysis was conducted using Python (Pandas, Matplotlib, and Seaborn libraries) and R (ggplot, tidyr, and dplyr libraries). Descriptive statistics were used to summarize distributions of responses across survey categories.

To evaluate relationships between categorical variables, Chi-Square tests of independence were performed. For paired pre- and post-intervention comparisons, the Wilcoxon signed-rank test was used due to the ordinal nature of the data. Due to the relatively small sample size and the presence of

sparse category counts, more complex models, such as logistic regression, were explored but not used as the primary basis for interpretation. Hypothesis testing was also conducted.

G. Ethical Considerations

Participation in the study was entirely voluntary and anonymous. No personally identifiable information was collected. Participants were informed of the purpose of the study and were free to discontinue at any time. No incentives were provided for participation.

IV. STUDY WORKFLOW

To address our research questions in a structured and reproducible manner, we designed a workflow that maps each research question (RQ) to specific survey variables. Rather than analyzing isolated survey responses, we grouped questions into conceptual constructs, such as awareness, risk perception, behavior, trust, and susceptibility. Each research question was then investigated using a targeted subset of these variables.

A. RQ1: "Do People Know About These Risks?" (Awareness)

To assess baseline awareness, we analyzed participants' responses to knowledge-based and perception-based questions related to AI data handling and system behavior. The following questions were used:

- Q5 and Q6: True/false questions assessing understanding of data storage and usage;
- Q8: Perception of AI output reliability;
- Q7: Self-reported confidence in understanding data flow;
- Q10: Open-ended responses describing perceived data handling;
- Q15: Confidence scale of risk perception;
- Q16: Trust assessment based on rank order.

These questions allowed us to evaluate both *objective awareness* (correctness of responses) and *perceived awareness* (self-reported confidence), and to identify potential misconceptions in participants' understanding of AI systems.

B. RQ2: "Do Users Feel at Risk?" (Risk Perception)

To evaluate perceived risk, we analyzed responses to a scenario-based question and related trust indicators:

- Q4: Self-reported frequency of sensitive data uploads to AI weekly;
- Q15: Perceived risk of uploading sensitive documents (e.g., tax-related data) to an AI assistant;
- Q14: Trust in AI systems to handle personal data securely;
- Q16: Ranking of most trusted entity for handling personal data (AI assistant vs. human professional vs. contextual).

Risk perception was primarily assessed through Q15, while Q4, Q14, and Q16 provided supporting context on users' trust and preferences. The relationship between trust and perceived risk was further examined to identify potential inconsistencies in user perception.

C. RQ3: "If Aware, Do They Care?" (Awareness vs. Behavior)

To investigate whether awareness translates into safer behavior, we analyzed participants' self-reported actions and usage patterns, and compared them with awareness indicators from RQ1. The following questions were used:

- Q1–Q4: Frequency and type of AI usage, including the frequency of sharing personal information;
- Q11: Types of information shared with AI systems (e.g., passwords, IDs);
- Q12: Their frequency of verifying AI-generated content;
- Q13: Likelihood of following AI-generated instructions without verification;
- Q14: Self-reported trust in AI systems.

These variables were analyzed to identify potential gaps between knowledge and behavior, particularly in cases where participants demonstrate awareness of risks but still engage in potentially unsafe practices.

D. RQ4: "Who Is Most Susceptible?" (Demographic Analysis)

To investigate susceptibility across different groups, we constructed a *misconception score* for each participant based on incorrect responses to key knowledge questions (Q5, Q6, Q8, and Q20). This score represents the number of misconceptions held by each participant. Since these questions are of the True/False type, each "False" answer was incremented to the individual's score.

Notice that a misconception score of 0 represents full awareness, while 4 represents complete misinformation, given the scope of the questions our participants were asked.

We then analyzed the distribution of this score across demographic variables, including:

- Age (Q0)
- Education level (Q0.1)

Due to the uneven distribution of participants across demographic categories, our results are interpreted descriptively rather than inferentially.

E. RQ5: "Does Education Help?" (Intervention Impact)

To evaluate the impact of our information-based intervention, we compared pre- and post-intervention responses related to verification behavior and risk-aware decision-making. The following paired questions were analyzed:

- Pre-intervention: Q12 and Q13
- Post-intervention: Q19–Q21

Because these responses are paired and ordinal, we used the Wilcoxon signed-rank test to assess whether observed changes were statistically significant. This analysis measures whether exposure to targeted educational content influences safer AI-use behavior.

Notice that the questions indicated above represent the majority of the variables we used to logically answer each of our research questions, although other complementary variables were also utilized to strengthen our analysis.

V. DATA ANALYSIS

The section analyzes the results of the survey we conducted and a statistical analysis of the findings. We aim to answer five research questions: whether users know about AI-related risks, whether they feel at risk, whether awareness affects behavior, which groups appear most susceptible to misconceptions, and whether a short educational intervention improves safer behavior.

A. RQ1: "Do People Know About These Risks?" (Awareness)

To begin, we analyzed whether users are aware of the risks of AI-related privacy and security risks, including their understanding of how AI systems store data, use submitted information, and generate responses. To do so, we asked respondents several questions regarding these matters. Fig. 1 presents the proportion of correct and incorrect responses that we collected.

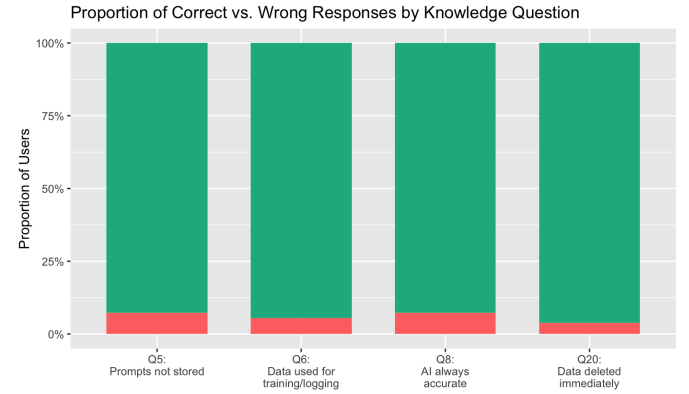


Fig. 1: Count of users who answered incorrectly for knowledge questions.

The data suggest that users are mostly aware of the risks of AI, but a few may lack knowledge of AI's accuracy and data handling. To further our investigation, we assigned a "misconception score" to users and tracked whether the same users answered multiple knowledge questions incorrectly.

TABLE I: Summarization of the amount of misconceptions of respondents throughout the knowledge-testing questions.

Misconception Score	Count	Proportion
0	45	73.8%
1	5	8.2%
2	4	6.6%
3	7	11.5%
4	0	0%

In our sample, 73.8% of respondents answered all questions correctly. From the few respondents who incorrectly answered questions, many answered one or two questions incorrectly. Because of this, we can infer that the majority of users have at least some knowledge of security regarding AI.

To further our investigation, respondents were asked two questions: "How confident are you that you understand where

your data goes after submitting it to an AI assistant?” and “How risky do you believe the following scenario is? Uploading tax-related documents to an AI assistant for analysis.” The results are visualized in Fig. 2 and Fig. 3.

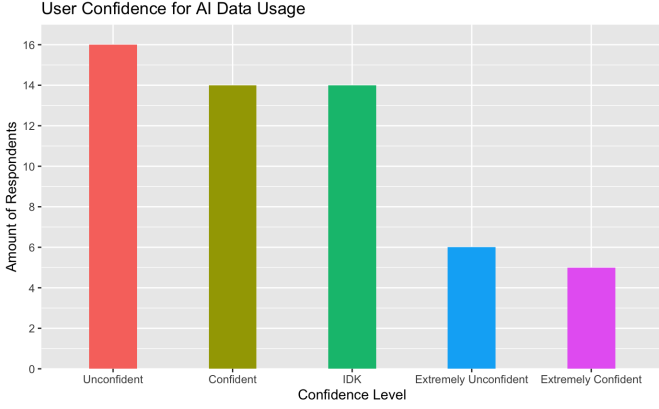


Fig. 2: Distribution of users’ responses to how confident they are that they know where their data is going after submitting it to AI.

Although 73.8% of the respondents answered the knowledge-based questions correctly, the large majority of respondents reported that they do not exactly know how AI is handling their data. From this, we can see that users have some perception of risk towards AI; however, they may not have a deep understanding of its inner workings.

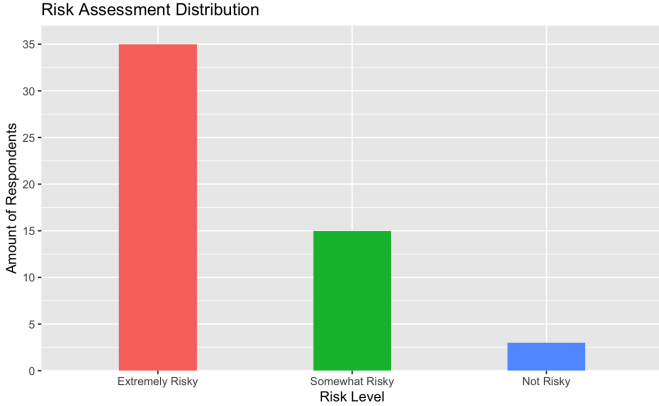


Fig. 3: Distribution of users’ risk perception towards uploading tax information into AI.

From the data, we can see that, again, the majority of respondents were able to perceive the tax-information scenario as risky. It further proves that users have a strong surface-level baseline of risks associated with AI, despite possibly not knowing exactly what the risks themselves are.

B. RQ2: “Do Users Feel at Risk?” (Risk Perception)

This subsection examines respondents’ trust in AI systems and their perceptions of privacy-related risk when using AI with sensitive information. Fig. 4 describes the surveyors’ trust

in AI. In our survey, we found that about about 78% of users had little to no trust in AI. Most of the data was clustered on the lower end of trust, while only three users claimed to trust AI a lot and completely.

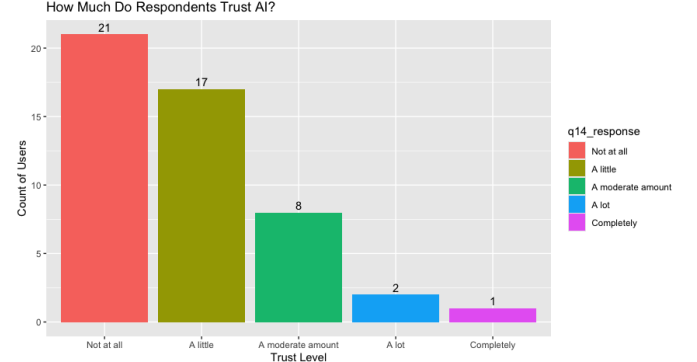


Fig. 4: Distribution of Respondents’ Trust in AI

To further our investigation, we then tested whether AI usage frequency was associated with trust in AI to see if frequent AI usage affected users’ risk perception. Fig. 5 shows that the majority of users, despite their AI usage frequency, claimed not to trust AI. The small number of respondents who reported high trust in AI was concentrated among those who reported more frequent AI use.

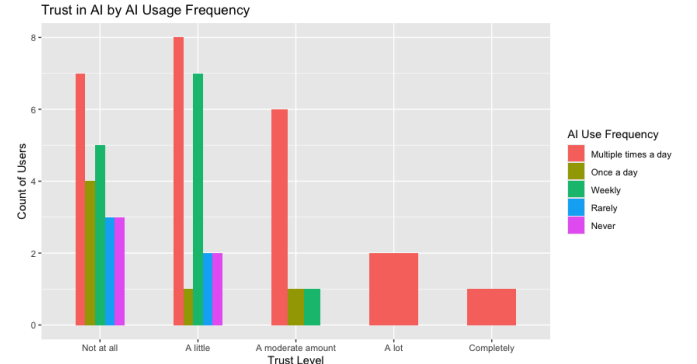


Fig. 5: Distribution of respondents’ trust in AI by their AI usage frequency.

To determine whether the AI usage frequency was associated with users’ trust in AI from Fig 5 showed any statistical association, a Chi-Square test was conducted. Table II summarizes the results.

TABLE II: Chi-square test results for AI usage frequency and trust in AI.

statistic	df	p_value
11.5615	16	0.7736

Using the Chi-Square test, we did not find a statistically significant association between AI usage frequency and trust in AI in this sample, as the P-value was .8084, which is well above .05. This suggests that users who reported using AI more often still feel as though AI has risks. Overall, the data

suggest that users are aware that using AI carries risks, and frequent AI usage does not alter whether their risk perception fluctuates.

C. RQ3: "If Aware, Do They Care?" (Awareness vs. Behavior)

In this subsection, we explore whether users' distrust of AI affects their data input behavior. In Table II, we concluded that frequent AI usage does not affect whether they trust AI. Users who frequently use AI are not more likely to distrust AI than those who do not frequently use it. However, despite this, the frequent usage may lead them to input sensitive documentation.

The respondents of our survey were asked to check off the following data they have shared with an AI: passwords/credentials, personal documents (resumes, IDs, etc.), work/school information, medical/legal information, or none. Asking such questions allowed us to classify which forms of data are the most common for users to leak.

TABLE III: Summary of users' behaviors when inputting sensitive information in AI.

Finding	Percent
Frequent AI users with low trust in AI	62.5%
Frequent AI users who input personal documents into AI	55%
Frequent AI users who input at least one sensitive info type	63%
Once-a-day users who input at least one sensitive info type	14%

Despite 62.5% of frequent AI users from Table III claiming they do not trust AI much, 55% of these frequent users also claimed to have input personal documents at least once to AI. We also found that 63% of frequent AI users have input at least one form of sensitive documentation into AI, compared to only 14% of once-a-day users performing the same behaviors. The data suggests that the AI usage of a user may affect whether they input certain forms of personal information. To strengthen the assumption, Table IV summarizes a Chi-Square test, testing whether there is a real association between the factors.

TABLE IV: Chi-Square test results for AI usage frequency and sharing data categories.

category	statistic	p_value	df
Passwords or credentials	1.2801	0.8647	4
Personal documents	14.0350	0.0072	4
Work/school confidential info	7.1766	0.1268	4
Medical/legal info	6.2945	0.1782	4
None	14.8243	0.0051	4

The chi-square analysis showed that the personal information and none categories were significantly associated with AI usage frequency, while the other categories were not. This suggests that the frequency of AI use may be related to some forms of information-sharing behavior. We also compared

reported trust in AI with the types of information users reported sharing to analyze whether users' risk perception affects specific forms of information they input into AI. To do so, we conducted a Chi-Square test, summarized in Table V, to determine whether there is an association.

TABLE V: Chi-Square test results for AI usage frequency and trust levels amongst respondents.

category	statistic	p_value	df
Passwords or credentials	18.2967	0.0011	4
Personal documents	15.6395	0.0035	4
Work/school confidential info	11.9900	0.0174	4
Medical/legal info	13.3702	0.0096	4
None	14.0300	0.0072	4

Using the Chi-Square test, we can see that all of the input categories are statistically significantly correlated with trust in AI. The p-values for all categories are significantly below .05, indicating that the user's trust level was linked to whether they entered this information into AI. In short, users' awareness of the risks of AI does indeed affect whether they input sensitive information into AI. Additionally, we found that users who frequently use AI are more likely to input personal documents than those who use AI less often.

D. RQ4: "Who Is Most Susceptible?" (Demographic Analysis)

We next analyzed the respondents' responses to knowledge questions related to AI data storage, data usage, and accuracy. These questions were used to identify possible misconceptions about how AI systems work.

We first determined which age range (Fig. 6) made up the proportion of respondents who incorrectly answered the knowledge questions.

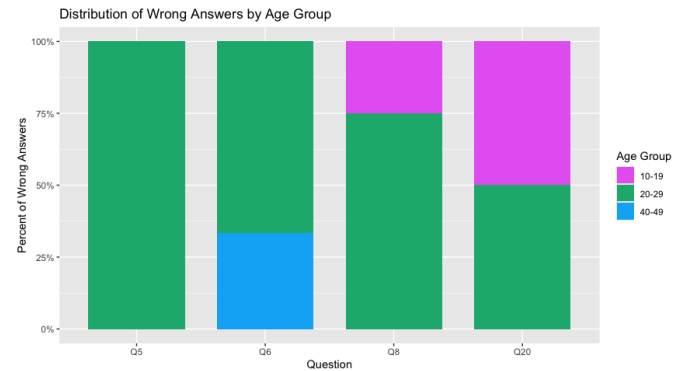


Fig. 6: Proportional distribution of respondents who answered incorrectly by age group.

We then determined which education level (Fig. 7) made up the proportion of respondents who incorrectly answered the knowledge questions.

The distribution of wrong answers was concentrated among respondents aged 20-29 and among respondents with college-level education. However, because these groups also made up a large share of the sample, this alone does not prove that

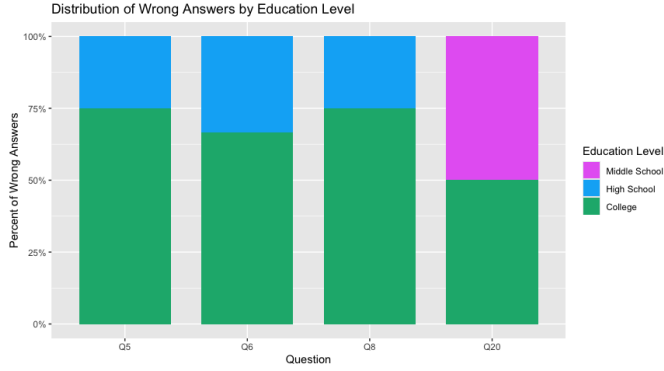


Fig. 7: Proportional distribution of respondents who answered incorrectly by education level.

they were inherently more susceptible from this sample. To strengthen our investigation, we revisited the misconception scores from Table I to see which age group and education level populated the results. The results are visualized in Fig. 8 for the education level.

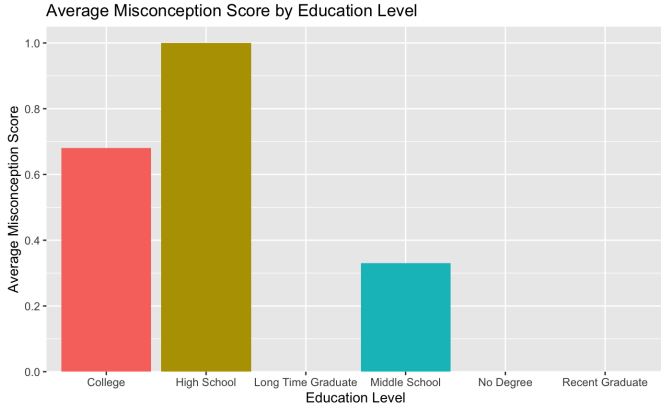


Fig. 8: Distribution of average misconception scores sorted by education level.

From this, we found that high school students and college students had the highest level of misconceptions compared to other education levels. Our respondents mostly consisted of high school and college students, with fewer long-time graduates, no degree, and recent graduates. Because of this, the data is naturally less populated towards these specific demographics. Similarly, we compared the misconception scores to the age groups, as shown in Fig. 9.

Like Fig. 8, the data for Fig. 9 consisted mostly of those between the ages of 20 and 29. Hence, naturally, that age range had the highest misconception score. Notably, age groups 10 to 19 and 40 to 49 had the same misconception score, indicating they have a similar amount of misconceptions on AI data usage. Overall, from the figures above, we can conclude that high school and college students, typically between the ages of 20 and 29, are the most susceptible to having misconceptions of AI.

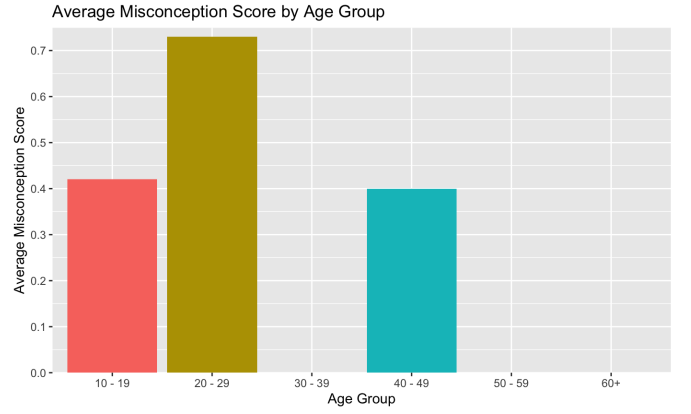


Fig. 9: Distribution of average misconception scores sorted by age groups.

E. RQ5: "Does Education Help?" (Intervention Impact)

Finally, we tested whether a short educational intervention changed how likely users were to verify AI-generated text and links before acting on them, as shown in Fig. 10.

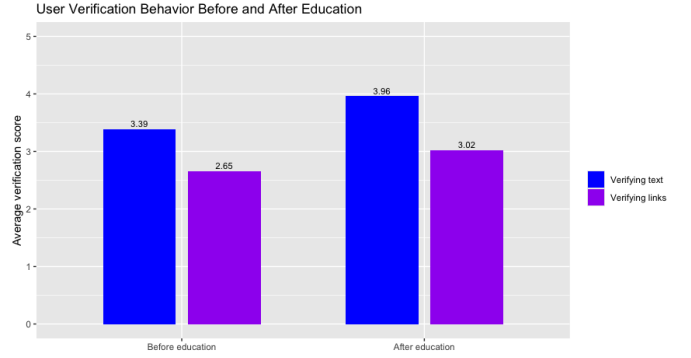


Fig. 10: Visualization of user verification behavior before and after education.

Visually, the appearance of the post-education bars for both verifying texts and verifying links is slightly higher. While analyzing only the averages, which are placed above the bars, we can see 3.39 shoot up to 3.96 and 2.65 shoot up to 3.02, post-education. However, to determine whether this change was statistically significant, we conducted a paired nonparametric test. Because these are paired before-and-after responses from the same participants, a paired Wilcoxon signed-rank test was used, as summarized in Table VI.

TABLE VI: Wilcoxon test results for respondents' answers to verifying text and links from AI before and after being educated.

measure	statistic	p_value
Verifying text	81.0	0.0157
Verifying links	37.5	0.0003

The educational intervention significantly improved reported verification behavior for both text and links. The change

was statistically significant for text and also statistically significant for links, with the link result showing especially strong evidence of improvement. These findings suggest that even a brief educational prompt may improve safer AI-use behavior.

VI. DISCUSSION

Our findings reveal a critical disconnect between user awareness and user behavior. While 73.8% of respondents answered all knowledge questions correctly, many still engaged in risky behaviors such as uploading sensitive documentation to AI assistants. This aligns with broader research showing that despite widespread adoption of generative AI tools, there remains a significant lack of user awareness regarding data collection and privacy practices [15]. Specifically, 62.5% of frequent AI users reported having little to no trust in AI, yet 55% of these same users admitted to inputting personal documents into AI systems at least once. This confirms that knowledge alone does not translate into safe behavior, and that users may operate under what we term "awareness without action."

A particularly striking finding was that most respondents reported having little to no trust in AI assistants, with approximately 78% of users clustered on the lower end of the trust scale. However, despite this low trust, 63% frequent users still shared sensitive information, while only 14% of once-a-day users inputted the same type of information. This shows that users who use AI frequently are more susceptible to inputting sensitive information into AI than those who use AI less frequently.

The Chi-square analysis revealed that AI usage frequency was not statistically associated with trust in AI ($p = 0.7736$), meaning that using AI more often does not make users more distrustful or more cautious. Instead, frequent users appear to normalize risky behavior over time, a finding that is particularly concerning given OpenAI's own research showing that approximately 30% of consumer ChatGPT usage is tied to work-related tasks, with millions of users feeding these systems commercially sensitive or personally identifiable information [16].

Our analysis of susceptibility revealed that high school and college students, typically between the ages of 20 and 29, are the most likely to hold misconceptions about AI data practices. This age group consistently had the highest misconception scores compared to other demographics. This finding is particularly concerning given that this demographic represents the heaviest users of AI assistants for academic and professional tasks. Research by Alshammari and Alshammari [15] similarly found that many users engage with AI platforms without fully understanding what data is being collected or the implications for their privacy, emphasizing that addressing these gaps is critical to ensuring responsible AI deployment.

The educational intervention proved effective. A brief informational prompt significantly improved reported verification behavior for both text ($p = 0.0157$) and links ($p = 0.0003$). This suggests that even low-effort educational interventions

can produce measurable improvements in safer AI-use behavior. However, the persistence of some misconceptions despite high overall awareness indicates that surface-level knowledge is insufficient. Users need not just facts, but also practical guidance and repeated reinforcement to change entrenched habits. This is consistent with Chen et al. [17], who found that while detection mechanisms for indirect prompt injection attacks are improving, post-processing mitigation remains an underexplored area requiring further attention.

VII. LIMITATIONS & FUTURE DIRECTIONS

This study has several limitations. First, our sample of 56 participants was recruited primarily from university student populations using convenience sampling. While this provided access to a high-frequency user group, it limits the generalizability of our findings to broader populations, including older adults, non-technical users, and corporate employees. Second, our data relied entirely on self-reported behaviors, which may be subject to social desirability bias or inaccurate recall. Third, the cross-sectional nature of our survey design captures attitudes and behaviors at a single point in time, preventing us from drawing causal conclusions about long-term behavior change. Fourth, our educational intervention measured immediate post-intervention responses rather than long-term retention or sustained behavior change.

Second, we will be focusing on 20-29 year old college students, who our data show are the most susceptible to misconceptions about AI data practices. This group requires targeted educational initiatives before they enter the workforce with established risky habits. Yi et al. [18] demonstrated that existing LLMs are universally vulnerable to indirect prompt injection attacks, and their proposed defense mechanisms, most of which are related to boundary awareness and explicit reminders, reduced attack success rates to near-zero levels. Translating these technical defenses into user-facing educational content represents a promising direction for future work.

Additionally, we recommend conducting more seminars about awareness of AI use and risks. Future work should evaluate the effectiveness of different educational formats, such as interactive workshops, video tutorials, and gamified learning modules, to determine which approaches produce the strongest and most lasting behavior change. Longitudinal studies are also needed to assess whether the improvements we observed immediately after intervention persist over weeks or months. Finally, researchers should investigate the organizational and systemic factors that enable or discourage risky AI use, including default privacy settings, warning messages, and institutional policies.

A critical area for future research is addressing the "confused deputy" problem in agentic AI systems [19]. As AI agents gain the ability to act, such as querying databases, calling APIs, and accessing files, the authorization boundary must shift from the model level to the tool level. Our findings suggest that users are largely unaware of these architectural vulnerabilities, and future work should investigate how to communicate these risks effectively to non-technical users.

VIII. CONCLUSIONS

This study investigated whether users overtrust AI assistants by examining their misconceptions, risk perceptions, and behaviors. Through a survey of 56 participants and an educational intervention, we made four key contributions. First, we found that while most users have surface-level awareness of AI risks, many hold specific misconceptions about data storage, deletion, and system accuracy. This aligns with broader findings that users' understanding of AI data practices remains significantly limited despite widespread adoption [15]. Second, we identified a significant gap between awareness and behavior: frequent users continued to share sensitive information even when they reported low trust in AI. Third, we determined that high school and college students aged 20 to 29 are the most susceptible population, with the highest misconception scores among all demographic groups. Fourth, we demonstrated that a brief educational intervention significantly improves verification behavior, particularly for AI-generated links.

Our findings have important implications for system designers, educators, and policymakers. System designers should implement stronger warnings before users upload sensitive information, particularly for tax documents, personal identifiers, and work-related files. Research by Yi et al. [18] suggests that defense mechanisms such as boundary awareness and explicit reminders can reduce attack success rates to near-zero levels while preserving output quality, and these same principles could be adapted for user-facing warnings. Educators should prioritize AI literacy programs targeting college students and young adults, focusing not just on facts but on practical safe usage behaviors. Employers should establish clear policies governing the use of AI assistants with company data, recognizing that employees may unknowingly expose confidential information.

The central takeaway of our research is this: awareness without action is not enough. Users may know the risks intellectually but still engage in unsafe behaviors due to habit, convenience, or a false sense of security. This mirrors the technical reality that even well-designed AI systems remain universally vulnerable to manipulation when proper authorization boundaries are not enforced [19]. However, education helps. Even a short intervention produced statistically significant improvements in safer behavior. By combining targeted education with better system design and organizational policies, we can help users move from awareness to action, reducing the privacy and security risks associated with the growing adoption of AI assistants.

IX. SURVEY

A. Demographics

- 1) *Q0*: How old are you? (MCQ - Select One)
- 2) *Q0.1*: What is your highest level of education? (MCQ - Select One)

B. AI Usage

- 1) *Q1*: How often do you use AI assistants (ChatGPT, Gemini, Copilot, Claude, etc.)? (MCQ - Select One)
- 2) *Q2*: What do you primarily use AI for? (MCQ - Select All)
- 3) *Q3*: Estimate how many times per week you use AI assistants. (Numeric)
- 4) *Q4*: Estimate how many times per week you input personal information into AI. (Numeric)

C. Knowledge & Misconceptions

- 1) *Q5*: AI systems do NOT store or use my prompts and information after I submit them. (True/False)
- 2) *Q6*: Information shared with AI can be used for AI training or logging purposes. (True/False)
- 3) *Q7*: How confident are you that you understand where your data goes after submitting it to an AI assistant? (Confidence Scale)
- 4) *Q8*: The information generated by AI is always accurate. (True/False)
- 5) *Q9*: Based on your answer for Q8 above, why or why not is "The information generated by AI is always accurate"? (Open-Ended)
- 6) *Q10*: In your own words, what do you think happens to your data and information after you submit it to an AI assistant? (Open-Ended)

D. Behavior

- 1) *Q11*: Which of the following have you ever shared with an AI? (MCQ - Select All)
- 2) *Q12*: How often do you verify AI-generated information (text) before using it? (Confidence Scale)
- 3) *Q13*: If an AI provides a link, instruction, or command, how likely are you to follow it without verification? (Confidence Scale: Scenario-Based)

E. Risk Perception & Trust

- 1) *Q14*: How much do you trust AI assistants to handle your data and personal information securely? (Confidence Scale)
- 2) *Q15*: How risky do you believe the following scenario is: "Uploading tax-related documents to an AI assistant for analysis" (Confidence Scale: Scenario-Based)
- 3) *Q16*: Who do you tend to trust the most with your data? (Rank Order)

F. Information-Based Intervention

G. Education Impact

- 1) *Q17*: After reading the information above, how often will you verify AI-generated information (text) before using it? (Confidence Scale)
- 2) *Q18*: Now, if an AI provides a link, instruction, or command, how likely are you to follow it without verification? (Confidence Scale)
- 3) *Q19*: Which of the following should you avoid sharing with an AI? (MCQ - Select All)

- 4) Q20: AI tools always delete your data immediately after each session. (True/False)
- 5) Q21: Which of the following is the safest way to use AI with sensitive information? (MCQ - Select All)

REFERENCES

- [1] E. Sandosham, "We don't understand ai at all," <https://futurism.com/anthropic-ceo-admits-ai-ignorance>, 2024, futurism, accessed 2026-04-12.
- [2] G. O'Brien *et al.*, "User misconceptions of llm-based conversational programming assistants," *arXiv preprint arXiv:2510.25662*, 2025. [Online]. Available: <https://arxiv.org/abs/2510.25662>
- [3] OpenAI, "Openai usage statistics," <https://openai.com>, 2024, accessed 2026-04-12.
- [4] P. Spencer, "Your employees are already using ai—with your company's confidential data," *Kiteworks*, 2025. [Online]. Available: <https://www.kiteworks.com/cybersecurity-risk-management/employees-sharing-confidential-data-unauthorized-ai-tools>
- [5] L. French, "38% of ai-using employees admit to sending sensitive work data," *SC Media*, 2024. [Online]. Available: <https://www.scworld.com/news/38-of-ai-using-employees-admit-to-sending-sensitive-work-data>
- [6] X. Zhan, J. Carrillo, W. Seymour, and J. Such, "Malicious llm-based conversational ai makes users reveal personal information," *USENIX Security Symposium*, 2025.
- [7] Check Point Software Technologies and Jerusalem Post, "Can you really trust your 'private' ai assistant to keep your secrets?" <https://www.jpost.com/>, 2026.
- [8] Z. Zhang, B. Guo, and T. Li, "Can humans oversee agents to prevent privacy leakage?" *arXiv*, vol. 2411.01344, 2025. [Online]. Available: <https://arxiv.org/abs/2411.01344>
- [9] E. Gumusel, Y. Yan, and E. Otenen, "From awareness to practice: A survey of u.s. users' privacy perceptions in llm chatbots," in *NDSS Symposium*, 2026.
- [10] L. Malki, A. Polamarasetty, M. Hatamian, M. Warner, and E. Costanza, "Hoovered up as a data point: Exploring privacy behaviours, awareness, and concerns among uk users of llm-based conversational agents," *PoPETs*, 2025.
- [11] Concentrix, "Why trust collapses when ai can't read the room," <https://www.concentrix.com/insights/blog/why-trust-collapses-when-ai-cant-read-the-room/>, 2025.
- [12] M. Yazan *et al.*, "Personality over precision: Exploring the influence of human-likeness on chatgpt use for search," *arXiv*, vol. 2511.06447, 2025. [Online]. Available: <https://arxiv.org/abs/2511.06447>
- [13] M. Cheng *et al.*, "Sycophantic ai decreases prosocial intentions and promotes dependence," *Science*, 2026.
- [14] Y. Yoon, "From trust to overreliance: Understanding the evolution of human-ai interactions and the role of ai explainability," Doctoral Dissertation, UMass Boston, 2025.
- [15] N. Alshammari and A. Alshammari, "Privacy concerns in chatgpt data collection and its impact on individuals," *Future Internet*, vol. 17, no. 11, p. 511, 2025.
- [16] D. Betti, "How people use chatgpt: Analysis of openai's first user study," *TechRadar*, 2026.
- [17] Y. Chen *et al.*, "Can indirect prompt injection attacks be detected and removed?" *arXiv preprint*, vol. 2502.16580, 2025. [Online]. Available: <https://arxiv.org/abs/2502.16580>
- [18] J. Yi, Y. Xie, B. Zhu, E. Kiciman, G. Sun, X. Xie, and F. Wu, "Benchmarking and defending against indirect prompt injection attacks on large language models," in *ACM Conference*, 2025.
- [19] QuarksLab, "Agentic ai: The confused deputy problem," 2026. [Online]. Available: <https://blog.quarkslab.com/agentic-ai-the-confused-deputy-problem.html>

X. INDIVIDUAL CONTRIBUTIONS

A. Raameen Ahmed

- Survey development and question formation
- Data analysis
- Paper writing (Data analysis)

- Presentation Development (Motivation & Contributions, Research Question & Methodology, Results & Insights: [Overall Results, Q2, Q3, Q5], Recommendations)

B. Pedro Callado de Paiva

- Survey development and question formation
- Data analysis
- Paper Formatting
- Paper writing (Abstract, Methodology, Study Workflow, Survey)
- Presentation development (Exploratory Data Analysis, Results & Insights: [Q1, Q4], Recommendations)

C. Muhammad Sheraz

- Paper writing (Background & Related Work, Discussion, Limitations & Future Directions, Conclusions)
- Presentation development (Recommendations, References)

D. Mohammed Alsheraa

- Paper writing (Introduction)
- Presentation development

E. Lea Nahli

- Paper writing (Abstract)
- Presentation development (Our Process, Overall Results, What Can We Do?, The Power of Education)